Phase 2 — Repo readiness vs interconnect taxonomy

Multi-user inference = **W11** (primary FP: continuous-batch TP under concurrent users). Realistic arrivals = **W21** (Azure/BurstGPT replay). MoE EP multi-user also hits **W12**. W07–W09 length grids are single-regime controls, not multi-user. Counts: smoke=9 · code=5 · partial=4 · missing=6.

Fabric caveat: VCF currently 1× h100-sxm-8-0 (+ 1×H100 NVL). Cross-node IB needs your 2nd node. Smoke NVLink first; IB when 2-node is up.

W01–W12

ID	Family	Role	Status	Start?	Repo anchors	Gap
W01	Dense DP / DDP train	train+	**smoke**	Y	ddp_resnet18/50_cifar10; run_workload_multi_gpu	1-node NVLink; IB 2-node untested
W02	FSDP / ZeRO-3 train	train+	**smoke**	Y	fsdp_{7,13,30,70}b_nvlink	Same
W03	TP + DP train	train+	**missing**	N	—	No Megatron / 3D train
W04	Pipeline 1F1B train	confuse	**missing**	N	—	No PP harness
W05	MoE EP train	train+	**missing**	N	moe_ep_synthetic = infer A2A only	Need real MoE train
W06	DP / replicate infer	infer-	**smoke**	Y	llm_infer_*; parallel replicas	Shared-queue multi-replica thin
W07	TP infer prefill-heavy	infer-	**smoke**	Y	qwen25_7b_prefill_heavy; mixtral_tp2_*	Not multi-user
W08	TP infer decode-heavy	infer-	**smoke**	Y	*_decode_heavy	Not multi-user
W09	TP infer midfill	infer-	**smoke**	Y	llm_midfill_*	Agentic loops thin
W10	Disagg KV P2P	FP	**missing**	N	ICTD kv_* not here	Biggest FP hole
W11	Heavy cont-batch TP serve	FP	**code**	Y	moe_serve_qwen7b_tp8_mixed; run_serving_trace --tp N	MULTI-USER PRIMARY
W12	MoE EP infer	FP	**code**	Y	moe_serve_*_ep8_*; run_moe_serving_vcf.sh	Needs 8 GPU + vLLM EP

W13–W24 (continued)

ID	Family	Role	Status	Start?	Repo anchors	Gap
W13	DiLoCo / local SGD	FN	**code**	Y	diloco_production.py; run_diloco_vcf.sh	Not in registry
W14	Streaming DiLoCo	FN	**code**	N	--streaming --num-fragments	Unsmoked
W15	Grad-accum / rare sync	FN	**partial**	Y	adversarial_N_*; K_ddp_accum_*; L_ddp_*	NVML-era; thin STEP logs
W16	LoRA / PEFT-only sync	FN	**partial**	N	lora_training; whitebox_lora_*	Mostly single-GPU
W17	PowerSGD / Top-k	FN	**missing**	N	—	No hook workload
W18	KV-shaped grad disguise	FN	**missing**	N	ICTD only	Port needed
W19	Hybrid duty-cycle	FN	**partial**	N	F_interleave; K; L diluted	Not wire-shaped hybrid
W20	Fake collectives on serve	FP	**code**	Y	moe_ep_synthetic_fake_ar; week7 EC1	Not on live vLLM yet
W21	Trace-replay serving	infer-	**code**	Y	serving_trace_replay; run_serving_trace	Not in registry; feeds W11
W22	Fine-tune / short SFT	confuse	**smoke**	Y	gpt2/bert; llm_train_*_lora	Label/regime issue
W23	Intra-node-only (IB quiet)	FN	**partial**	Y	Any 1-node ddp/fsdp	No dual NVLink/IB collector
W24	Idle / non-ML	neg	**smoke**	Y	idle; cufft/nbody/mining/render	—

First-session smoke order

1 idle · 2 ddp_resnet18_cifar10 · 3 fsdp_7b_nvlink · 4 qwen25_7b prefill+decode · 5 moe_serve_qwen7b_tp8_mixed **or** run_serving_trace (W11/W21 multi-user) · 6 diloco via run_diloco_vcf.sh H=500 · 7 moe_ep_synthetic_fake_ar · 8 optional moe_serve_olmoe_ep8_decode.

Build next (block research claim)

FP: W10 disagg KV (GB-scale P2P) — without it adversarial FPR understated. **FN:** register DiLoCo W13/W14; DDP-LoRA-only W16; port ICTD kv_disguise W18; PowerSGD W17. W15 grad-accum already usable stand-in for rare sync.

Colors: green=smoke · blue=code · amber=partial · red=missing. SPAR Fall 2026.